higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING
NQF LEVEL 2

18 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 C
1.2 D
1.3 B
1.4 C
1.5 B

(5 × 1) **[5]**

QUESTION 2

2.1 2.1.1 Elevation of A = 1 000 + 1 600 – 350 ✓
= 2 250✓
Elevation of B = 1 000 + 1 600 – 850✓
= 1 750✓ (4)

2.1.2 Height difference between A and C = 850 – 350✓
= 500 ✓ (2)

2.1.3 The difference in reading when viewed along two different lines of sight (1)

2.2
- For accurate marking and multiple cutting of the same lengths
- Prevents the tape from getting in the way, being bent or damaged when taking measurements
- Allows the user to mark a specific point on the tape
- The tape can be locked to mark the end of a measurement
- Keeps the blade in position
- Saves time (Any 3 × 1) (3)

2.3
- Tape measure
- Spirit level
- Steel pegs and stakes
- Hammer and nails
- Builders line
- Builders square
- Profile boards
- Plumb bob
- Raging rods (Any 5 × 1) (5)

2.4
- It can only be used for small buildings
- As the triangle is made of wood, its size and shape can change
- It will break if dropped
- It is not as accurate as measuring diagonally
- Difficult to use on uneven surfaces (Any 3 × 1) (3)

2.5
- Measure both the diagonals that cross each other
- Both measurements must be the same length (2)

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2.6
- Set up the theodolite at point A closer to the building
- Set out a 90° angle to fix point B
- Set up the theodolite at B and zero circle onto point A
- Measure the vertical angle ø to the point to be measured
- Vertical distance = $\frac{AB}{Tan\ ø}$ ✓

(5)
[25]

QUESTION 3: FOUNDATIONS

3.1 It is a floating foundation✓ that makes a building look like a boat.✓ (2)

3.2
- It consists of a box-like rigid structure set at a certain depth below ground.✓ The weight of the soil removed to place it on✓ equals the weight of the building.✓
- It is buoyed up✓ by the weight of the earth✓ displaced in creating the foundation.✓ (Any 1 × 3) (3)

3.3 It is used on wet soil✓ and when the soil is soft,✓ the friction piles will not support the load of the building.✓ (3)

3.4 Thickness = 200 mm ✓
Width = 600 mm ✓ (2)
[10]

QUESTION 4: BEHAVIOUR OF STRUCTURES

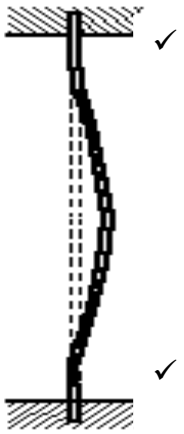
4.1
4.1.1 Compression
4.1.2 Strain
4.1.3 Strain
4.1.4 Neutral plane
4.1.5 Shear load

(5 × 1) (5)

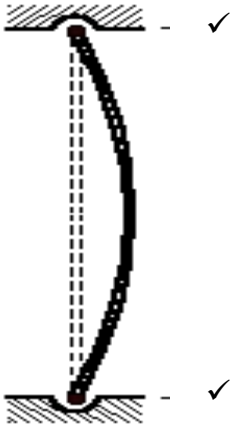
4.2 Four-point failure:✓ When an eccentric load is applied,✓ the arch starts buckling at different places.✓
Five-point failure:✓ When a central load is applied,✓ the sides flare out and breakage of the sides occurs.✓ (6)

4.3
- A pin-joint works like a lapped joint.
- It transfers vertical and horizontal shear loads and
- it cannot resist any bending or moment (rotational) forces. (Any 2 × 1) (2)

4.4



BOTH ENDS FIXED



BOTH ENDS PINNED (4)

4.5 4.5.1 An equivalent load✓ applied to a single point✓

4.5.2 It is a distributed load, which has a constant value✓ along the length.✓

4.5.3 It is a load of which the magnitude varies along the loading length✓ at a constant rate.✓

4.5.4
- A force administered✓ along the lines of an axis✓
- When an object is loaded so that the force is normal✓ to the fixed axis✓

(Any 1 × 1)
(4 × 2) (8)
[25]

QUESTION 5: INTERNATIONAL SYSTEM (SI) UNITS OF MEASUREMENTS

5.1
5.1.1 D
5.1.2 A
5.1.3 B
5.1.4 E
5.1.5 C

(5 × 1) (5)

5.2
5.2.1 True
5.2.2 False
5.2.3 True
5.2.4 False
5.2.5 False

(5 × 1) (5)
[10]

QUESTION 6: AREAS AND VOLUME

6.1 Total number of painters needed = $\frac{5\ 000}{250}$ ✓

= 20 ✓

Number of painters for each day = $\frac{20}{5}$

= 4 painters ✓ (3)

6.2 Total length = (5 000 × 9) ✓+ (6 000 × 5) ✓ + (220 × 11)✓

= 45 000 + 30 000 + 2 420 ✓

= 77 420 mm ✓

Volume (converted to meters) = 77,42 × 0,6 × 0,2✓

Volume = 9,29 m³✓ (7)

[10]

QUESTION 7: TECHNICAL DRAWINGS

7.1 7.1.1 Storm water drain
7.1.2 Polyvinyl chloride
7.1.3 Mild steel
7.1.4 Natural ground level
7.1.5 Water closet

(5 × 1) (5)

7.2 •To check that the drawing meets certain requirements
•To avoid problems that could be encountered during the building process (2)

7.3 A – Plaster
B – Lintel
C – Glass
D – DPC
E – Sill (5)

7.4 7.4.1
- An orthographic projection – looking down on the object, such as in a floor plan.
- A section viewed from the top. In such views, the portion of the object above the plane is omitted to reveal what lies beyond.
- View of an object as projected on a horizontal plane.
- The appearance of an object as seen from above. (Any 1 × 1)

7.4.2
- A view used on a drawing to show an area or hidden part of an object by cutting away or removing some of that object.
- Portrays a cut-away portion of the object and is another way to show hidden components in a plan.
- Sliced view of a part or component in a drawing showing its internal make-up. (Any 1 × 1)

7.4.3
- Orthographic projection drawing that shows one side of a house.
- Shows the finished appearance of a given side of the house and furnishes vertical height dimensions.
- A horizontal orthographic projection of a building onto a vertical plane, the vertical plane normally being parallel to one side of the building.
- Two-dimensional pictures of the façade of a house. (Any 1 × 1)

(3 × 1) (3)

[15]

TOTAL: 100

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